

CHAPTER 1:

Exercise 1.1

Let A , B and C be sets.

1. If $a \notin A \setminus B$ and $a \in A$, show that $a \in B$.

In other words...

prove: $a \notin A \setminus B \wedge a \in A \Rightarrow a \in B$

NOTE: it will be demonstrated by left-to-right transit

$$a \notin A \setminus B \wedge a \in A$$

$\equiv$ < Rewriting of $\notin$ >

$$\neg(a \in A \setminus B) \wedge a \in A$$

$\equiv$ < Definition of $\setminus$ >

$$\neg(a \in A \wedge a \notin B) \wedge a \in A$$

$\equiv$ < De Morgan's law >

$$(\neg(a \in A) \vee \neg(a \notin B)) \wedge a \in A$$

$\equiv$ < Rewriting of $\notin$ >

$$(a \notin A \vee \neg(\neg(a \in B))) \wedge a \in A$$

$\equiv$ < Double negation LAW >

$$(a \notin A \vee a \in B) \wedge a \in A$$

$\equiv$ < Distributive Laws >

$$(a \notin A \wedge a \in A) \vee (a \in B \wedge a \in A)$$

$\equiv$ < Contradiction for $\wedge$ >

$$\text{false} \vee (a \in B \wedge a \in A)$$

$\equiv$ < Neutral element for $\wedge$ >

$$a \in B \wedge a \in A$$

$\Rightarrow$ < Debilitation >

$$a \in B$$



2. Show that if $A \subseteq B$, then $C \setminus B \subseteq C \setminus A$

In other words...

prove: $A \subseteq B \Rightarrow (C \setminus B \subseteq C \setminus A)$

Proof: $A \subseteq B \Rightarrow (C \setminus B \subseteq C \setminus A)$

$\equiv$ < Definition of $\subseteq$ >

$$(\forall x \mid x \in A: x \in B) \Rightarrow (C \setminus B \subseteq C \setminus A)$$

$\equiv$ < Other way >

$$(\forall x \mid x \in A \Rightarrow x \in B) \Rightarrow (C \setminus B \subseteq C \setminus A)$$

$\equiv$ < Contrapositive LAW >

$$(\forall x \mid x \notin B \Rightarrow x \notin A) \Rightarrow (C \setminus B \subseteq C \setminus A)$$

Note: by the deduction Metatheorem it is enough to demonstrate...

$$C \setminus B \subseteq C \setminus A$$

$\equiv$ < definition of $\subseteq$ >

$$(\forall x \mid x \in C \setminus B: x \in C \setminus A)$$

$\equiv$ < Definition of $\setminus$ >

$$(\forall x \mid x \in C \wedge x \notin B: x \in C \wedge x \notin A)$$

$\Leftarrow$ < Hypothesis >

$$(\forall x \mid x \in C \wedge x \notin A: x \in C \wedge x \notin A)$$

$\equiv$ < Reflexive >

true



3. Suppose $A \setminus B \subseteq C$. Show that $A \setminus C \subseteq B$

In other words...

Prove: $A \setminus B \subseteq C \Rightarrow A \setminus C \subseteq B$

Proof:

$$A \setminus B \subseteq C \Rightarrow A \setminus C \subseteq B$$

$\equiv$ < Definition of $\subseteq$ >

$$(\forall x | x \in A \setminus B : x \in C) \Rightarrow A \setminus C \subseteq B$$

$\equiv$ < Definition of $\setminus$ and other form >

$$(\forall x | : x \in A \wedge x \notin B \Rightarrow x \in C) \Rightarrow A \setminus C \subseteq B$$

$\equiv$ < Contrapositive Law >

$$(\forall x | : x \notin C \Rightarrow \neg(x \in A \wedge x \notin B)) \Rightarrow A \setminus C \subseteq B$$

$\equiv$ < De Morgan's Law >

$$(\forall x | : x \notin C \Rightarrow x \notin A \vee x \in B) \Rightarrow A \setminus C \subseteq B$$

Note: by the deduction Metatheorem it is enough to demonstrate...

$$A \setminus C \subseteq B$$

$\equiv$ < Definition of $\subseteq$ >

$$(\forall x | x \in A \setminus C : x \in B)$$

$\equiv$ < Definition 1 >

$$(\forall x | x \in A \wedge x \notin C : x \in B)$$

$\Leftarrow$ < Hypothesis >

$$(\forall x | x \in A \wedge (x \notin A \vee x \in B) : x \in B)$$

$\equiv$ < Distributive Law >

$$(\forall x | (x \in A \wedge x \notin A) \vee (x \in A \wedge x \in B) : x \in B)$$

$\equiv$ < Contradiction, Neutral element of $\vee$ >

$$(\forall x | x \in A \wedge x \in B : x \in B)$$

$\equiv$ < Other form, Debititation >

true



4. Suppose $A \subseteq B$ and $A \subseteq C$. Show that $A \subseteq B \cap C$

In other words...

prove: $A \subseteq B \wedge A \subseteq C \Rightarrow A \subseteq B \cap C$

proof:

Note: by the deduction Metatheorem it is enough to demonstrate...

$$A \subseteq B \cap C$$

$\equiv$ < definition of $\subseteq$ >

$$(\forall x | x \in A : x \in B \cap C)$$

$\equiv$ < Definition of $\cap$ >

$$(\forall x | x \in A : x \in B \wedge x \in C)$$

$\equiv$ < $\quad \quad \quad$ >

$$(\forall x | x \in A : x \in B) \wedge (\forall x | x \in A : x \in C)$$

$\equiv$ < Definition of $\subseteq$ >

$$A \subseteq B \wedge A \subseteq C$$

$\equiv$ < Hypothesis >

true



5. Suppose $A \subseteq B$ and $B \cap C = \emptyset$. Show that $A \subseteq B \setminus C$

In other words...

Prove:

$$A \subseteq B \wedge B \cap C = \emptyset \Rightarrow A \subseteq B \setminus C$$

NOTE: it will be demonstrated by left-to-right transit

$$A \subseteq B \wedge B \cap C = \emptyset$$

$$\equiv \langle \text{lemma} \rangle$$

$$A \subseteq B \wedge B = B \setminus C$$

$$\equiv \langle \text{Substitution} \rangle$$

$$A \subseteq B \setminus C$$



Lemma:

$$B \cap C = \emptyset \equiv B = B \setminus C$$

NOTE: it will be demonstrated by left-to-right transit

$$(\forall x) : x \in B \wedge x \notin C \equiv \text{false})$$

$$\equiv \langle$$

$$(\forall x) : x \notin B \vee x \notin C)$$

$$\equiv \langle$$

$$(\forall x) : x \in B \Rightarrow x \notin C)$$

$$\equiv \langle$$

$$(\forall x) : x \in B \equiv x \in B \wedge x \notin C)$$

$$\equiv \langle$$

$$B = B \setminus C$$



6. Show that $A \setminus (B \setminus C) \subseteq (A \setminus B) \cup C$

In other words...

prove:

$$A \setminus (B \setminus C) \subseteq (A \setminus B) \cup C$$

proof:

$\Rightarrow$

$$(\forall x \mid x \in A \setminus (B \setminus C) : x \in (A \setminus B) \cup C)$$

$\Rightarrow$

$$(\forall x \mid x \in A \setminus (B \setminus C) \Rightarrow x \in (A \setminus B) \cup C)$$

MTG

$$x \in A \setminus (B \setminus C) \Rightarrow x \in (A \setminus B) \cup C$$

$$x \in A \wedge x \notin (B \setminus C)$$

$\Rightarrow$

$$x \in A \wedge (x \notin B \vee x \in C)$$

$\Rightarrow$

$$(x \in A \wedge x \notin B) \vee (x \in A \wedge x \in C)$$

$$x \in A \setminus B \vee x \in C$$

$\Rightarrow$

$$(x \in A \wedge x \notin B) \vee x \in C$$

$\Rightarrow$

$$(x \in A \vee x \in C) \wedge (x \notin B \vee x \in C)$$

$$= x \in A \vee x \in C$$

$\Leftarrow$

$$x \in A$$

$$= \text{true}$$

7. Show that $A \setminus B \subseteq C$ and $A \not\subseteq C$, then $A \cap B \neq \emptyset$

In other words...

prove: $A \setminus B \subseteq C \wedge A \not\subseteq C \Rightarrow A \cap B \neq \emptyset$

Proof:

Lemma

$$A \cap B = \emptyset \equiv A \setminus B = A$$

NOTE: It will demonstrate by contradiction. ($\neg$ false $\equiv$ true)

$$\neg (A \setminus B \subseteq C \wedge A \not\subseteq C \Rightarrow A \cap B \neq \emptyset)$$

$\equiv$ < Definition of $\Rightarrow$, De Morgan Laws >

$$A \setminus B \subseteq C \wedge A \not\subseteq C \wedge A \cap B = \emptyset$$

$\equiv$ < Lemma >

$$A \setminus B \subseteq C \wedge A \not\subseteq C \wedge A = A \setminus B$$

$\equiv$ < Substitution >

$$A \subseteq C \wedge A \not\subseteq C$$

$\equiv$ < Contradiction >

False



Exercise 2.1

Prove the following theorems, where A, B, C , and D are sets:

1. Theorem. If $A \subseteq B$, then $A \subseteq A \cup B$ and $A \cap B \subseteq A$.

In other words...

Prove:

$$\underline{A \subseteq B \Rightarrow A \subseteq A \cup B \wedge A \cap B \subseteq A}$$

Proof:

$$A \subseteq B \Rightarrow A \subseteq A \cup B \wedge A \cap B \subseteq A$$

$\equiv <$

$$(A \subseteq B \Rightarrow A \subseteq A \cup B) \wedge (A \subseteq B \Rightarrow A \cap B \subseteq A)$$

Note: First, it will demonstrate $A \subseteq B \Rightarrow A \subseteq A \cup B$

$$(A \subseteq B \Rightarrow A \subseteq A \cup B)$$

$$(\forall x | x \in A : x \in A \vee x \in B)$$

$\equiv <$

$$(\forall x | : x \in A \Rightarrow x \in A \vee x \in B)$$

$\equiv <$

$$(\forall x | : x \notin A \vee (x \in A \vee x \in B))$$

$\equiv <$

$$(\forall x | : \text{true})$$

$\equiv <$

true